# Toward Optimal Homotopy Paths for Multivariate Systems via Reinforcement Learning: a progressive approach through Pham systems

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**Abstract**

This document presents a progressive approach to studying Pham systems of increasing complexity, with the ultimate goal of applying reinforcement learning (RL) to design optimal homotopy paths for numerical stability in polynomial system solving. We focus on systems where start and target share critical properties (zero-dimensionality, common monomial bases) to avoid numerical difficulties.

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# 1 Introduction

## 1.1 Project Context

In a previous work, we have addressed homotopy continuation between systems composed of a single univariate polynomial. This document propose a roadmap to extend that work to some more complex systems in a context where one can maintain the following constraints:

- **Zero-dimensional systems**: Only systems with a finite number of solutions are considered.
- **Common monomial basis**: The quotient rings of the start and target systems must share a common monomial basis.
- **Numerical stability**: The condition number must remain manageable throughout the homotopy path and the homotopy path must avoid the singular locus.

## 1.2 Basic presentation of Pham systems

**Example 1.1** (Pham system)**.** A Pham system is a multivariate polynomial system of the form:

$$f_1 = x_1^{d_1} + h_1, \ldots, f_n = x_n^{d_n} + h_n$$

where $d_1, \ldots, d_n \in \mathbb{Z}_{>0}$ and $h_1, \ldots, h_n \in K[x_1, \ldots, x_n]$ such that for any $i$, the total degree of $h_i$ is $< \min(d_1, \ldots, d_n)$.

**Lemma 1.2.** *With the same notations, such a presentation of a Pham system $F = (f_1, \ldots, f_n)$ is a Gröbner basis for grevlex, grlex or any degree-refining monomial ordering. In addition, the ideal $\langle F \rangle$ is of dimension zero.*

*Proof.* Let us assume $\leq_\omega$ is grevlex, grlex or any degree-refining monomial ordering. Then

$$\mathtt{lm}_{\leq_\omega}(f_i) = x_i^{d_i}.$$

As such, applying Buchberger's product criterion and Buchberger's theorem, we get that for any $j$, $(f_1, \ldots, f_j)$ is a Gröbner basis for $\leq_\omega$ . This is in particular true for $i = n$.

We now prove that $F$ is a regular sequence, meaning that for any $i$, $f_i$ is not a zero divisor in $K[x_1, \ldots, x_n]/\langle f_1, \ldots, f_{i-1} \rangle$ . Proceeding by contradiction, let us assume that $f_i$ is a zero divisor in $K[x_1, \ldots, x_n]/\langle f_1, \ldots, f_{i-1} \rangle$ . It means there is some $b \in K[x_1, \ldots, x_n]$ such that $bf_i \in \langle f_1, \ldots, f_{i-1} \rangle$ .

Compatibility with degree and products imply that $\mathtt{lm}(bf_i) = mx_i^{d_i}$ for some monomial $m \in M$. However, $\langle \mathtt{lm}(f_1), \ldots, \mathtt{lm}(f_{i-1}) \rangle \subset K[x_1, \ldots, x_{i-1}]$. This is a contradiction and we then gets that $(f_1, \ldots, f_n)$ is a regular sequence. The dimension of an ideal generated by a regular sequence of length $n$ in $K[x_1, \ldots, x_n]$ is zero, which concludes the proof. □

## 1.3 Condition number?

Homotopy continuation is a powerful method for solving polynomial systems by continuously deforming a known system (the **start system**) into a target system. The **condition number** should be a measure of the sensitivity of the system to perturbations and thus be critical for ensuring numerical stability.

As such, there is no easy-to-present, easy-to-compute definition of the condition number of a square polynomial system.

One reasonable way to define a local condition number at a root $\xi$ of $F$ is:

$$\kappa(F, \xi) := \|JF_\xi\| \|JF_\xi^{-1}\|$$

where $JF_\xi$ is the Jacobian matrix of $F$ at $\xi$. It is indeed the condition number of $JF_\xi$ that should govern the local convergence of the Newton method for finding an update of $\xi$ as a root of a small perturbation of $F$.

A global condition number for $F$ (assumed to be of dimension zero) could then be:

$$\kappa(F) := \max_{\xi \in Z(F)} \kappa(F, \xi).$$

More refined results are available thanks to the works of Shub and Smale.

# 2 First Case: Fake Bivariate Pham Systems

## 2.1 System Definition

Consider a **Fake Bivariate Pham system** of the following form:

$$FBP(x_1, x_2) = \begin{pmatrix} f_1(x_1) \\ f_2(x_2) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{1}$$

where $f_1(x_1)$ and $f_2(x_2)$ are **univariate polynomials**.

## 2.2 Properties

### 2.2.1 Roots and Solutions

Let $\mathcal{Z}(f_1) = \{\zeta_1, \ldots, \zeta_{d_1}\}$ be the set of roots of $f_1$ (assumed simple), and $\mathcal{Z}(f_2) = \{\xi_1, \ldots, \xi_{d_2}\}$ be the set of roots of $f_2$ (also assumed simple).

**Theorem 2.1.** *The solutions of system (1) are:*

$$\left\{ \begin{pmatrix} \zeta_i \\ \xi_j \end{pmatrix} : i \in [\![1, d_1]\!], j \in [\![1, d_2]\!] \right\}$$

*This yields $d_1 \cdot d_2$ roots, reaching the **Bézout bound**.*

### 2.2.2 Quotient Ring Structure

The quotient ring $\mathbb{C}[x_1, x_2]/I$ has a basis as a $\mathbb{C}$-vector space:

$$B = \left\{ x_1^i \cdot x_2^j : i \in [0, d_1 - 1], j \in [0, d_2 - 1] \right\} \tag{2}$$

This allows the decomposition:

$$\mathbb{C}[x_1, x_2]/I = \bigoplus_{i \in [0, d_1 - 1]} \mathbb{C} \cdot x_1^i \cdot x_2^j = \bigoplus_{\boldsymbol{\alpha} \in B} \mathbb{C} \cdot \boldsymbol{x}^{\boldsymbol{\alpha}}$$

# 3 Second Case: General bivariate Pham Systems

## 3.1 System Definition

Consider a more general **bivariate Pham system**:

$$GBP(x_1, x_2) = \begin{pmatrix} x_1^{d_1} + f_1(x_1, x_2) \\ x_2^{d_2} + f_2(x_1, x_2) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{3}$$

where:

$$\max\{\deg(f_1), \deg(f_2)\} < \min\{d_1, d_2\}$$

### 3.2 Properties

#### 3.2.1 Gröbner Basis

As stated in Lemma 1.2, $GBP$ is a Gröbner basis for any degree-refining monomial ordering.

#### 3.2.2 Quotient Ring Structure

The quotient ring $\mathbb{C}[x_1, x_2]/I$ has the same basis as a $\mathbb{C}$-vector space as the analogue false bivariate system of same degrees:

$$B = \left\{ x_1^i \cdot x_2^j : i \in [0, d_1 - 1], j \in [0, d_2 - 1] \right\} \tag{4}$$

This allows the decomposition:

$$\mathbb{C}[x_1, x_2]/I = \bigoplus_{i \in [0, d_1 - 1]} \mathbb{C} \cdot x_1^i \cdot x_2^j = \bigoplus_{\boldsymbol{\alpha} \in B} \mathbb{C} \cdot \boldsymbol{x}^{\boldsymbol{\alpha}}$$

## 4 Reinforcement Learning for Homotopy Paths

### 4.1 Motivation

Traditional homotopy methods rely on **fixed paths**, which may not be optimal for numerical stability. Reinforcement learning (RL) offers a data-driven approach to learn optimal paths.

### 4.2 Objective

Find a good homotopy path between $FBP$ and $GBP$.

#### 4.2.1 Reward Function?

- Timing?
- Number of steps?
- Condition number along the path?

## 5 Further Generalizations

### 5.1 Higher-Dimensional Systems

Extend to $n$-variate Pham systems ($n > 2$): going from Fake n-variate Pham system to General n-variate Pham system.

### 5.2 More General Polynomial Systems

Consider non-Pham systems that are still zero-dimensional: shape position? Triangular decomposition?

# A Notation

| Symbol | Description |
|---|---|
| $FBP(x_1, x_2)$ | a Fake Bivariate Pham system $(f_1(x_1), f_2(x_2))$. |
| $GBP(x_1, x_2)$ | a General Bivariate Pham system $(x_1^{d_1} + f_1(x_1, x_2), x_2^{d_2} + f_2(x_1, x_2))$. |
| $\mathcal{Z}(f)$ | Set of roots of polynomial $f$ |
| $\Delta$ | Discriminant of a system |
| $\text{Res}(f, g)$ | Resultant of polynomials $f$ and $g$ |
| $J_F$ | Jacobian matrix of system $F$ |
| $\mathbb{C}[x_1, x_2]/I$ | Quotient ring by ideal $I$ |
| $\partial$ | Partial derivative operator |

Table 1: Notation used in this document.

# References


[1] Pham, F. (1969). *Singularités des systèmes différentiels de Gauss-Manin*. Bures-sur-Yvette: Institut des Hautes Études Scientifiques.

[2] Verschelde, J. (1999). *Newton's method and homotopy continuation for solving systems of polynomial equations*. Journal of Computational and Applied Mathematics.

[3] Sutton, R. S., *et al.* (2018). *Reinforcement Learning: An Introduction*. MIT Press.